

CPSC 304 Project Cover Page

Milestone #: 1

Date: February 4, 2026

Group Number: 107

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Domain: image capture and analysis

2a) Our project combines computer vision and image capture to translate sign-language into words. We want to be able to take in either a video or live footage and translate the alphabet from American sign language to English text using machine learning and databases. The domain also relates somewhat to accessibility and language translation and documentation.

2b) Our database models the following:

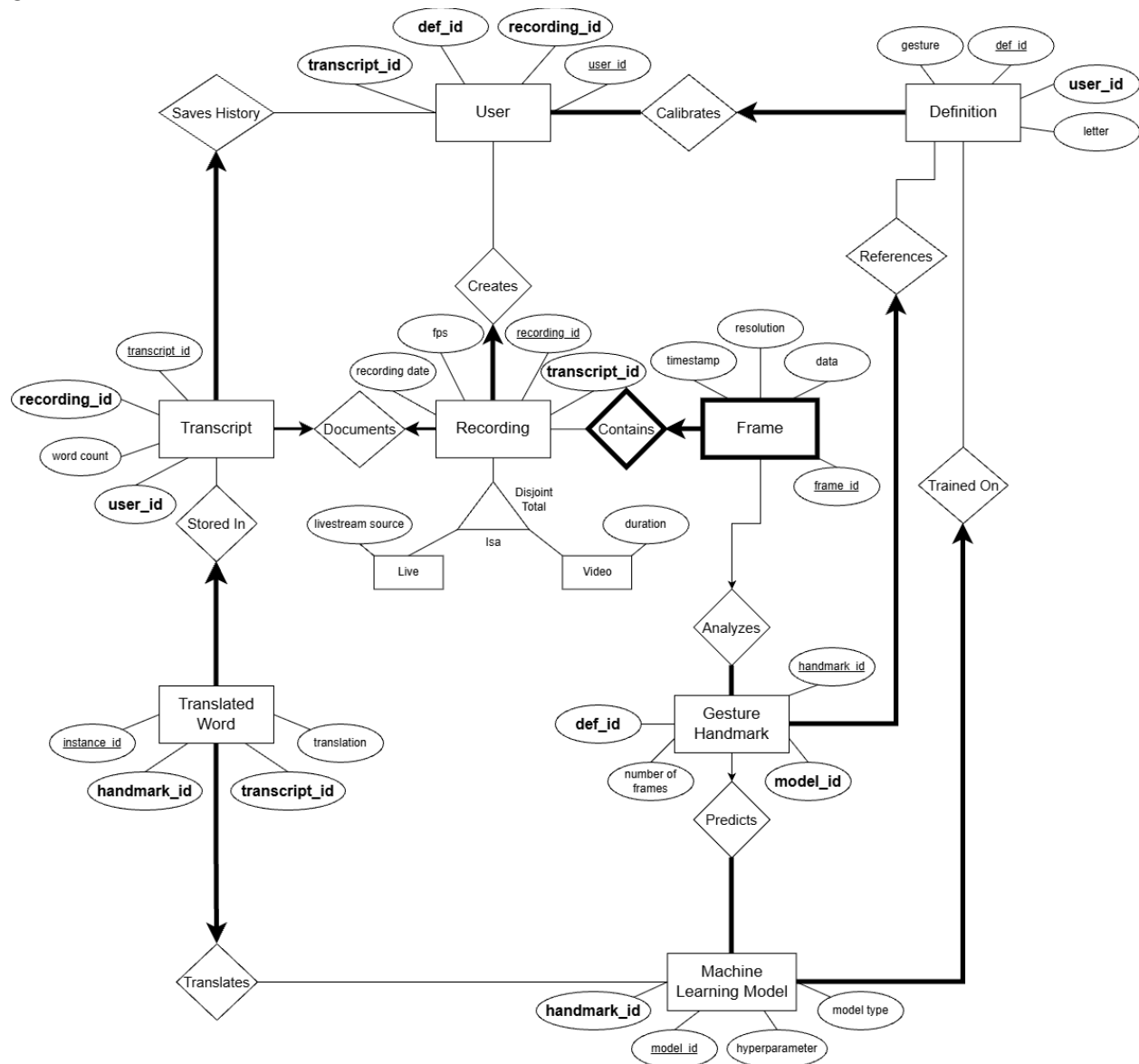
- **Users** who create recordings
- Video **Recordings** broken into **Frames** for analysis
- **Hand landmark** features extracted from images of the frames (shape, position, movement)
- A **Definition** of a known sign language symbol (alphabet letter) distinguished by their movement patterns.
- The **Model** is trained on definitions, and predicts the translation of detected gestures
- A **Translated Word** where the prediction of a particular gesture is stored
- The final **Transcript** generated from recognized signs

This allows our database to perform the following functions:

- Store user recordings and their history
- Allow users to create their own sign language definitions
- Organize video data into frames for analysis
- Save detected hand landmark features to compare to the dictionary
- Inputs gestures as features to predict English letters using a machine learning model
- Generate and store transcripts of video recordings

Our goal for this project is to make communication between ASL and non-ASL speakers easier without either party needing to compromise on their preferred language, starting with basic letters.

3.



Note: We have a lot of foreign keys that are bold, but it might be difficult to see. For foreign keys, the bubbles and text are bigger to make a clearer distinction. Thank you!

5. Additional info

To better understand the project, the following information will describe each entity and how they would practically be used.

User: The user will take recordings of themselves to store in the dictionary so that the machine learning model can be calibrated to the user's specific hand motions. The user must save at least one hand motion definition so that the model can use it to predict later recordings. Transcripts of recordings will be saved and tied to the user.

Definition: The definition is the feature data used to train the machine learning model and is comprised of a hand gesture and its associated meaning.

Gesture Handmark: The gesture handmark is the hand position, motion, and shape observed across several frames in the recording and is used for prediction by the model.

Machine Learning Model: The model trained on the user created definitions.

Translated Word: The final English word prediction made by the machine learning model after processing the given gesture.

Transcript: A documentation of all the translated words in a single recording session.

Recording: Recordings contain the weak entity "Frame", where a specific frame cannot be identified without the full recording. Additionally, recordings can be either live recordings or pre-recorded videos. There can be one transcript describing one entire recording.

6. We did not use AI for any part of this assignment.